AYUSH KUMAR AGARWAL

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PROFESSIONAL SUMMARY

Aspiring **Software Development Engineer** with a passion for developing **scalable, intelligent systems** across **Full-Stack Development**, **AI**, and **Machine Learning**. Enthusiastic about the intersection of technology and finance, with a keen interest in **quantitative analysis**, **risk assessment**, and **data-driven investment insights**. Committed to enhancing problem-solving skills to evolve as a versatile and impact-driven engineer.

WORK EXPERIENCE

HackNex Season 2 | Top 10 Finalist | AI Engineer & Lead Developer

Sep 2026

JIS College of Engineering, Kalyani

- Developed **FinOS**, a multi-agent AI financial intelligence platform integrating payslip analysis, salary breakdowns, tax insights, investment research, portfolio management, fraud detection, and risk analysis.
- Designed collaborative AI agent workflows using LangGraph, RAG/GraphRAG, and event-driven architecture to connect financial data and enable context-aware analysis.
- Focused on generating unified, explainable financial insights through shared financial knowledge, cross-agent collaboration, and evidence-based decision support.

Technologies / Skills Used : *Python, LangGraph, LLMs, RAG, GraphRAG, MCP, FastAPI, Neo4j, Qdrant, PostgreSQL, Apache Kafka*

HACKFORGE (Finalists) | Co-Developer

Apr 2026

Jadavpur University, Kolkata

- Co-developed **VeriFund**, a blockchain-based disaster relief platform promoting transparent and accountable fund distribution.
- Contributed to AI-powered features, frontend development, and backend integration to support core application functionality.
- Collaborated with the team on application logic, API integration, and UI/UX improvements to refine the overall product experience.

Technologies / Skills Used : *AI Integration, React, Node.js, API Integration, Frontend Development, Backend Development, UI/UX*

KidGuides (Hult Prize) | Founding Team & Product Developer

Feb 2026

Heritage Institute, Kolkata

- Co-developed an offline-first EdTech platform connecting school students with college mentors for accessible academic guidance.
- Contributed to product strategy, UI/UX design, and feature planning to improve the user experience.
- Conducted market research and real-world validation to identify user needs and support early-stage business development.

Technologies / Skills Used : *Product Development, UI/UX, Business Strategy, Market Research*

EDUCATION

HERITAGE INSTITUE OF TECHNOLOGY | Kolkata

Jun 2024 - Jun 2028

B.TECH IN COMPUTER SCIENCE AND BUSINESS SYSTEM

CGPA : 8.94

PROJECTS

FinOS | Multi-Agent AI Financial Intelligence Platform | Github

Sep 2026

AI Engineer & Lead Developer

- Built FinOS, a multi-agent AI financial intelligence platform using Python, LLMs, LangGraph, and MCP for coordinated financial analysis.
- Implementing RAG/GraphRAG, Neo4j, Qdrant, FastAPI, and Kafka for context-aware retrieval and event-driven financial insights.

Technologies / Tools Used : *Python, TypeScript, Next.js, React, FastAPI, PostgreSQL, SQLAlchemy, Redis, LangGraph, LLMs, MCP, RAG/GraphRAG, Neo4j, Qdrant, Apache Kafka, WebSockets*

JANSEWA | Github Website

Jul 2026

Full-Stack Developer

- Developed a full-stack civic-tech platform using React, Node.js, and MongoDB to help citizens discover relevant government welfare schemes.
- Implemented location-aware grievance reporting, AI integration, and complaint-tracking workflows with transparent status updates.

Technologies / Tools Used : *React, Vite, Node.js, Express.js, MongoDB, Mongoose, Groq AI, Leaflet, JWT, bcryptjs, node-cron, SHA-256, Chart.js / Recharts*

RESUMATCH | Github Jun 2026

AI & Full Stack Developer

- Developed an AI-powered ATS resume analysis platform using NLP and semantic similarity to evaluate resume compatibility with job descriptions and identify missing skills and keywords.
- Implemented ATS scoring, personalized improvement recommendations, analysis history, and downloadable PDF reports to streamline resume optimization.

Technologies / Tools Used : *Python, Streamlit, FastAPI, spaCy, Sentence Transformers, Groq API, Llama 3, Supabase, Google OAuth, Jinja2, WeasyPrint*

SMARTATTEND | Github Website May 2026

AI & Full Stack Developer

- Developed an AI-powered attendance management system using face and voice recognition to automate student attendance verification.
- Integrated classroom photo/audio processing with an AI pipeline and real-time cloud synchronization using Supabase and PostgreSQL.

Technologies / Tools Used : *Python, Streamlit, dlib, scikit-learn (SVM), Resemblyzer, librosa, Supabase, PostgreSQL, bcrypt, NumPy, Pandas, Pillow, QR Code*

OPEN-SOURCE

AstrBot | Open Source Contributon | Github Oct 2026 - Present

Open Source Software Contributor

- Improved remote help response validation with safeguards for malformed payloads, HTTP errors, and oversized responses.
- Added unit tests covering invalid responses, network failures, timeouts, and response-size limits.
- Contributed changes integrated into AstrBot's master branch through a maintainer-reviewed integration PR.

Technologies / Tools Used : *Python, Git, GitHub, REST APIs, JSON, Input Validation, Exception Handling, Debugging, Code Review, Pull Request Workflows*

CERTIFICATIONS

Python for AI/ML | Jan 2026 - May 2026

Apna College

- Completed the Python for AI/ML course from Apna College, gaining foundational knowledge of artificial intelligence and machine learning concepts, data-driven problem-solving, and modern AI applications through structured learning and practical exercises.

Technologies / Tools Used : *Python, NumPy, Pandas, Data Preprocessing, Machine Learning, Scikit-learn, Deep Learning, PyTorch, NLP, Generative AI, RAG, AI Agents, OpenAI API, Flask, SQL, Git, GitHub*

Financial Freedom Course | Dec 2025 - Mar 2026

Batch 15 Sapphire | Study2Win

- Completed coursework in financial literacy, stock market fundamentals, fundamental analysis, risk management, and investment strategies.

SKILLS

Programming Languages :	C, C++, Java, Python, JavaScript, TypeScript
Web Technologies :	HTML5, CSS3, React.js, Next.js, Node.js, Express.js
Frameworks & Libraries :	NumPy, Pandas, Scikit-learn, PyTorch, OpenCV, Streamlit, Flask, FastAPI
AI Technology:	LLMs, GenAI, RAG, AI Agents, NLP, Prompt Engineering, LangChain, LangGraph, Semantic Search
Tools & Platforms :	Git, GitHub, Postman, VS Code, Docker, Figma, Render, Vite
Databases :	MySQL, MongoDB, Supabase, Firebase
Soft Skills :	Analytical Thinking, Team Collaboration, Adaptability, Quick Learning
Languages :	English, Hindi, Bengali, Marwari